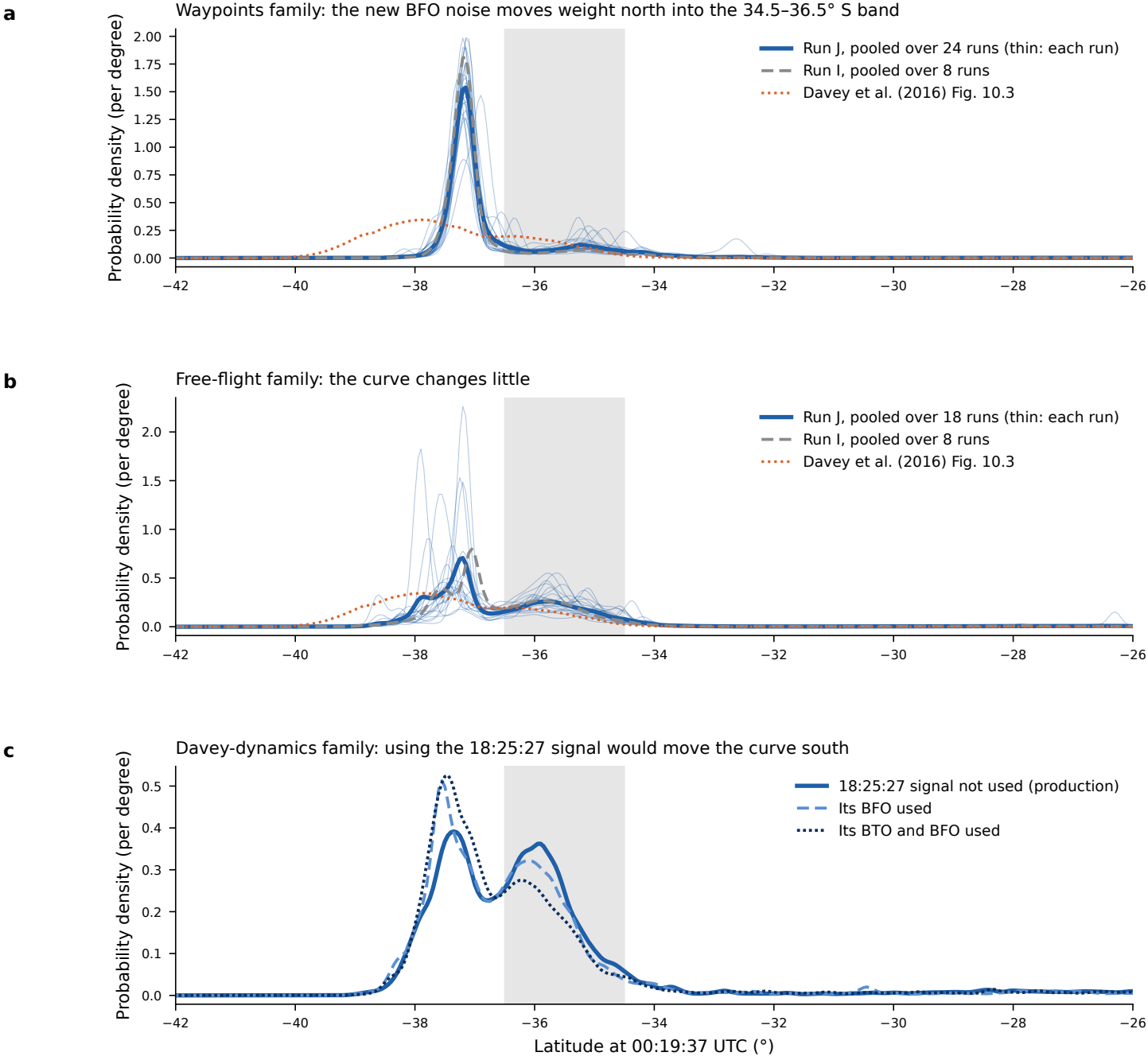


Plain (ASD-STE100): These curves show where the aircraft was at 00:19:37 UTC, by latitude. Run J uses a BFO noise of 4.32 Hz; run I used 7 Hz. Panel a: in the waypoints family, more probability is now between 34.5° S and 36.5° S (grey band): 0.167, before 0.109. The median moves 0.05°. Panel b: in the free-flight family, the curve changes little. Panel c: the 18:25:27 signal is not used in run J. If we use it, the curve moves about 0.5° south. The signal fits the other data well. The project lead accepts run J as converged (16 October 2026). North of 33° S, few independent paths carry the curves. The one-engine physics and the fuel-flow factor on extrapolated fuel-table steps are provisional.

Technical: core run J, tag run-j = 1244b7c, deskstar x86_64. Weighted particle histograms, 0.1° Gaussian smoothing, latitude of the final state (00:19:37, 7th arc). a: run-j-waypoints 24 × 7M (ln Z −127.82) vs next-i-waypoints 8 × 7M (BFO sd 7 Hz, no kappa_x); shoulder 0.109 → 0.167; median −37.149 → −37.097; overlap 0.89. b: run-j-free 18 × 7M (ln Z −134.49) vs next-i-free 8 × 7M; shoulder 0.372 → 0.374; overlap 0.85. c: run-j-repro-radar seeds 1–8 (3.5M each) vs companions run-j-repro-r600bfo / run-j-repro-r600 (18:25:27 R600 BFO 142 Hz sd 4.3177 Hz; BTO 12,520 µs sd 62), matched seeds; median −36.394 → −36.618 → −36.849, shoulder 0.445 → 0.395 → 0.336; held-out ln p(BFO) −2.95 ± 0.05, ln p(BTO | BFO) −5.76 ± 0.07 (≈0.9σ, 1.2σ). Production stack: BFO sd 4.3177 Hz (Davey Table 5.1); 18:25:27 held out; kappa_x ~ N(1.0, 0.046) PROVISIONAL; one-engine s7 + s8 hold-taper PROVISIONAL; 00:19 core treatment both BTO, no BFO; radar 18:04–18:22; fuel internal-v1, two tanks. Convergence: PASS by project ruling (Pete, 16 Oct 2026); all-partition split-half waypoints 0.934, free 0.873, repro 0.921. Northern lineages north of 33° S (r-weighted): waypoints 120, free 81. Grey band: Davey shoulder 34.5–36.5° S. Built 16 Oct 2026.